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Every Rivet and Every Bolt is Important

As a former Navy helicopter mechanic, I went into Ed Pierson's presentation expecting to hear about defective planes and why government oversight of private corporations is important. I was not expecting for the talk to feel so familiar. Mr. Pierson reminded me of several excellent officers that I had the pleasure of working for in the Navy, and struck me as a person that I would want as a leader. Additionally, I worked on or around systems like the ones Mr. Pierson showed and described in his slideshow. I know what it means to have to account for every part in thousands of parts, and how it feels to know everything about how those parts are installed.

What made me sad and angry about Mr. Pierson's explanation of Boeing's actions was the reasoning behind the normalization of deviance. Employees at every level were involved in the deviance, but ultimately the decisions were made by management for the sake of profits. Mr. Pierson explained how the "lag" introduced by supply chain engine shortages led to the "catch-up" of final inspections being squeezed versions of entire blocks of in-process inspections. I personally experienced the effects of similar leadership decisions where "how something is supposed to be done" was turned into "how something is actually done" for the sake of saving a couple hours. I do not blame the workers, I blame the managers who knew intimately how many shortcuts were being made on the assembly line but chose numbers over safety.

Mr. Pierson's passion for understanding the details of the situation was apparent. I was very happy to have so much background knowledge on the topic and felt pride as Mr. Pierson explained that the individuals building Boeing's planes are hardworking technicians. I felt one of

his statements deeply, it was something to the effect of, “every rivet and every bolt on an airplane is important.” This statement is also reminiscent of the Challenger disaster. Redundancy in flight systems is critical because the question is not if a part will fail, but when it will fail, and subsequently what can be done to ensure the people on that craft will make it safely to their destination. I feel very strongly about the reliability of safety systems because of this idea.

To a certain extent, I feel like a trusted confidant after listening to Mr. Pierson. His attention to detail on Boeing industrial systems, the parts of a plane that passengers don’t see, and the lack of transparency during investigations gave the audience an intense understanding of the situation. His ability to relay his knowledge and experiences so thoroughly made clear his whistleblower status as “hiding in plain sight.” Now that I have background knowledge about the fatal and near-accidents, I care deeply that Boeing is able to recover as a trustworthy company and that the victims’ families are able to get closure.

Main Takeaways

- Ed Pierson reminded me of my best officers from Naval aviation.
- Engineers don’t build things. They design them.
- Certain departments of the US government are enabling Boeing’s lack of transparency.
- Don’t be afraid to fly, instead understand that transparency is a requirement for safety.

Lingering Questions

- One of the Q&A questions didn’t have time for a full answer: What were the engineering decisions that contributed to the 737 Max crashes?
- What public or private organizations exist that can hold the US government accountable for failing to hold Boeing accountable? Is litigation just too expensive?